VULNERABILITY ASSESSMENT REPORT

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Website

scanme.nmap.org

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INTERNSHIP

FUTURE INTERNS - CYBER SECURITY

1. INTRODUCTION

A vulnerability assessment was conducted on scanme.nmap.org, a website designed for testing security tools. The assessment was performed using Nmap to identify open ports and the services running on them.
The scan revealed six open ports associated with different services. Two of these services are running older software versions, which may present potential security risks. The remaining ports are open but have limited identifiable information, requiring further monitoring.
Although no high-risk vulnerabilities were detected, the presence of medium-risk issues highlights the need for proper security measures. This report outlines the findings and provides practical recommendations to improve system security.

2. TOOLS USED

- Nmap (Network scanning tool)
- Browser Developer Tools

3. SCAN RESULTS

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Command Prompt

Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sheyi>nmap -v
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-21 02:05 +0200
Read data files from: C:\Program Files (x86)\Nmap
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.05 seconds
           Raw packets sent: 0 (0B) | Rcvd: 0 (0B)

C:\Users\sheyi>nmap -sV scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-21 02:06 +0200
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.29s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 994 closed tcp ports (reset)
PORT      STATE SERVICE    VERSION
22/tcp    open  ssh        OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http       Apache httpd 2.4.7 ((Ubuntu))
2000/tcp  open  tcpwrapped
5060/tcp  open  tcpwrapped
9929/tcp  open  nping-echo Nping echo
31337/tcp open  tcpwrapped
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/
Nmap done: 1 IP address (1 host up) scanned in 22.03 seconds

C:\Users\sheyi>
```

The Nmap scan was run against scanme.nmap.org using the -sV flag for version detection. The target was live at IP address 45.33.32.156 with a latency of 0.29 seconds. The scan completed in 22.03 seconds and found six open TCP ports. Port 22 runs SSH version 6.6.1p1, port 80 runs Apache 2.4.7, and port 9929 runs nping-echo. Ports 2000, 5000, and 31337 are open but marked as tcpwrapped. The remaining 994 ports were closed and returned a TCP reset packet.

4. VULNERABILITIES IDENTIFIED

-Scan Target: scanme.nmap.org (45.33.32.156)
-Open Ports and Analysis:

Port 22 (SSH) - OPEN
-Service: OpenSSH 6.6.1p1 Ubuntu
-Risk Level: Medium
Description: This port allows remote access to the server. The installed SSH version is outdated and may have known security holes. Attackers could try brute-force attacks to guess passwords and gain access.

Port 80 (HTTP) - OPEN
-Service: Apache httpd 2.4.7 (Ubuntu)
-Risk Level: Medium
Description: This port serves web traffic but uses HTTP instead of HTTPS. Any data sent between the user and the server is not encrypted. This makes it easy for attackers to steal information or perform Man-in-the-Middle (MITM) attacks.

Port 2000 (tcpwrapped) - OPEN
-Risk Level: Low
Description: This port is open but the service could not be fully identified. It may be protected by a TCP wrapper. The unknown nature means it should still be checked.

Port 5000 (tcpwrapped) - OPEN
-Risk Level: Low
Description: Similar to port 2000, this port is open but the service is not clearly visible. This port is sometimes used for development or testing services.

Port 9929 (nping-echo) - OPEN
-Service: Nping echo
-Risk Level: Low
Description: This service replies back with whatever data is sent to it. It is meant for network testing but could be misused by attackers to probe or flood the server.

Port 31337 (tcpwrapped) - OPEN
-Risk Level: Low
Description: This port is open but wrapped. The port number is often associated with "elite" backdoor tools. While likely just a test service here, it still needs monitoring.

5. RISK CLASSIFICATION

-High Risk – Critical vulnerabilities that could allow immediate system compromise or unauthorized access.

-Medium Risk – Vulnerabilities that may be exploited if left unaddressed and could lead to security breaches.

-Low Risk – Minor issues that pose limited risk but should still be monitored and managed.

6. RECOMMENDATIONS

For Port 22 (SSH) - Medium Risk

-Update OpenSSH from version 6.6.1p1 to the latest stable version
-Disable root login over SSH
-Use strong passwords or switch to key-based authentication
-Change the default SSH port from 22 to a non-standard port (optional)
Implement failed login attempt limits to stop brute-force attacks.

For Port 80 (HTTP) - Medium Risk

-Install an SSL/TLS certificate and switch from HTTP to HTTPS
-Redirect all HTTP traffic to HTTPS automatically
-Update Apache from version 2.4.7 to the latest version
-Remove any unnecessary Apache modules to reduce attack surface
-Enable security headers (HSTS, X-Frame-Options, etc.)

For Ports 2000, 5060, 31337 (tcpwrapped) - Low Risk

-Investigate what services are actually running on these ports
-Close any port that does not serve a legitimate business need
-Document all open ports for future security audits
-Add firewall rules to restrict access to trusted IPs only

For Port 9929 (nping-echo) - Low Risk

-Disable nping-echo service if not actively needed for testing
-If required, restrict access to authorized users only
-Monitor this port for unusual traffic patterns

7. CONCLUSION

The vulnerability assessment conducted on scanme.nmap.org identified six open ports on the target system. No high-risk or critical vulnerabilities were detected during the scan. However, two medium-risk issues were observed, mainly related to outdated SSH and Apache services.

The remaining four ports were classified as low risk but still require monitoring and further investigation.
Implementing security measures such as keeping software up to date and transitioning from HTTP to HTTPS would address the main concerns identified.

Additionally, performing regular security scans can help detect new vulnerabilities before they become serious threats. Overall, the system is reasonably secure but requires a few important improvements to strengthen its security posture.